Portfolio Page: Customer Payment Behavior and Risk

🔹 **Title:** Customer Payment Behavior and Risk: A Business Intelligence Approach

---

🔹 Introduction:

In this case study, I analyze customer payment behavior to identify patterns that signal increased financial risk — especially loan default. Acting as a business intelligence analyst for a national financial services company, I developed actionable insights to reduce repayment risk, support strategic decision-making, and improve operational performance.

Business leaders in fintech and lending sectors increasingly rely on KPIs and BI dashboards to assess loan health and monitor customer behavior. This case study simulates that real-world challenge, leveraging historical lending data to uncover risk indicators and drive smarter interventions.

🔹 Problem Statement:

Certain customer segments are defaulting at a higher rate, which threatens profitability and sustainability. The organization lacks visibility into which borrowers are most at risk.
Goal: Identify and monitor high-risk customer segments using clear, scalable BI tools.

🔹 Key Business Questions:

- What patterns exist in repayment and default behavior?
- Which financial indicators best predict loan default?
- How can the organization act early to reduce risk?

🔹 Dataset and Tools:

- **Data Source**: Cleaned Lending Club data (2018+)
- **Tools**:
  - **R**: Data cleaning, feature engineering, KPI creation
  - **Google Sheets**: Data review
  - **Tableau**: Dashboard development

- **GitHub**: Version control and documentation

🔗 View Tableau Dashboard
(Add link when published to Tableau Public)

🔹 Data Preparation:

- Cleaned missing values and removed irrelevant columns
- Standardized income and loan purpose categories
- Engineered features:
  - **Repayment-to-Income Ratio (RIR)**
  - **Income Tier**
  - **High-Risk Flag**

🔹 Analysis Methods:

- Aggregated default rates by income tier, loan purpose, and loan grade
- Created KPIs:
  - Default Rate
  - Avg. DTI
  - % of High-Risk Customers
- Visualized relationships using bar charts and heatmaps

📊 Example Visuals:

- Default Rate by Loan Purpose
- Heatmap: Risk by Income Tier & Grade
  (Add charts from your `visuals/` folder here)

🔹 Key Findings:

- Customers with:
    - Income < $40,000
    - DTI > 35%
    - Loans for debt consolidation
    - Grades E, F, or G
      ...were most likely to default.
- High **RIR** strongly correlated with missed payments

🔹 Solutions:

Primary: Deploy a real-time BI dashboard to flag risky customer segments
✅ Pros: Low-cost, scalable, promotes proactive decisions
⚠️ Cons: Requires data pipeline maintenance and user training

Alternative 1: Tighten loan approval criteria for at-risk profiles
Alternative 2: Offer customer financial education or counseling

---

🔹 Recommendations:

1. Build and launch Tableau dashboard using developed KPIs
2. Assign a BI team to monitor and update dashboard monthly
3. Train finance, operations, and support teams to use insights
4. Timeline: Internal test in 30 days, full rollout in 60

🔹 Reflection:

This case study strengthened my:

- Data wrangling and feature engineering skills (in R)
- Ability to design clear, decision-ready KPIs
- Dashboard creation and stakeholder-focused communication
- Understanding of how analytics solves real business challenges

---

🔹 Project Files (Structure Overview):

From your project folder `customer-payment-risk-bi/`:

plaintext
CopyEdit

```
📁 data/
  ├── raw/ (e.g., LendingClub.csv)
  └── cleaned/ (e.g., loans_2018_2020_cleaned.csv)

📁 scripts/
  └── clean_transform.R

📁 visuals/
  ├── heatmap_income_grade.png
  └── bar_rir_purpose.png

📁 dashboard/
  └── tableau_link.txt

📁 docs/
  └── Customer_Risk_Case_Study.pdf

📄 README.md (Project summary)
```

🔹 About Me:

Hi, I'm Kira — a data analyst-in-training and former educator based in Charleston, SC. I specialize in turning messy datasets into strategic insights, and I'm passionate about finance, business intelligence, and helping companies solve real problems with data.

📬 Connect with me on LinkedIn
📁 Download My Resume